Computer Networks

计算机网络（CST2398）

2025-2026学年第一学期（26-Spring）

计算机科学与技术学院 王洁

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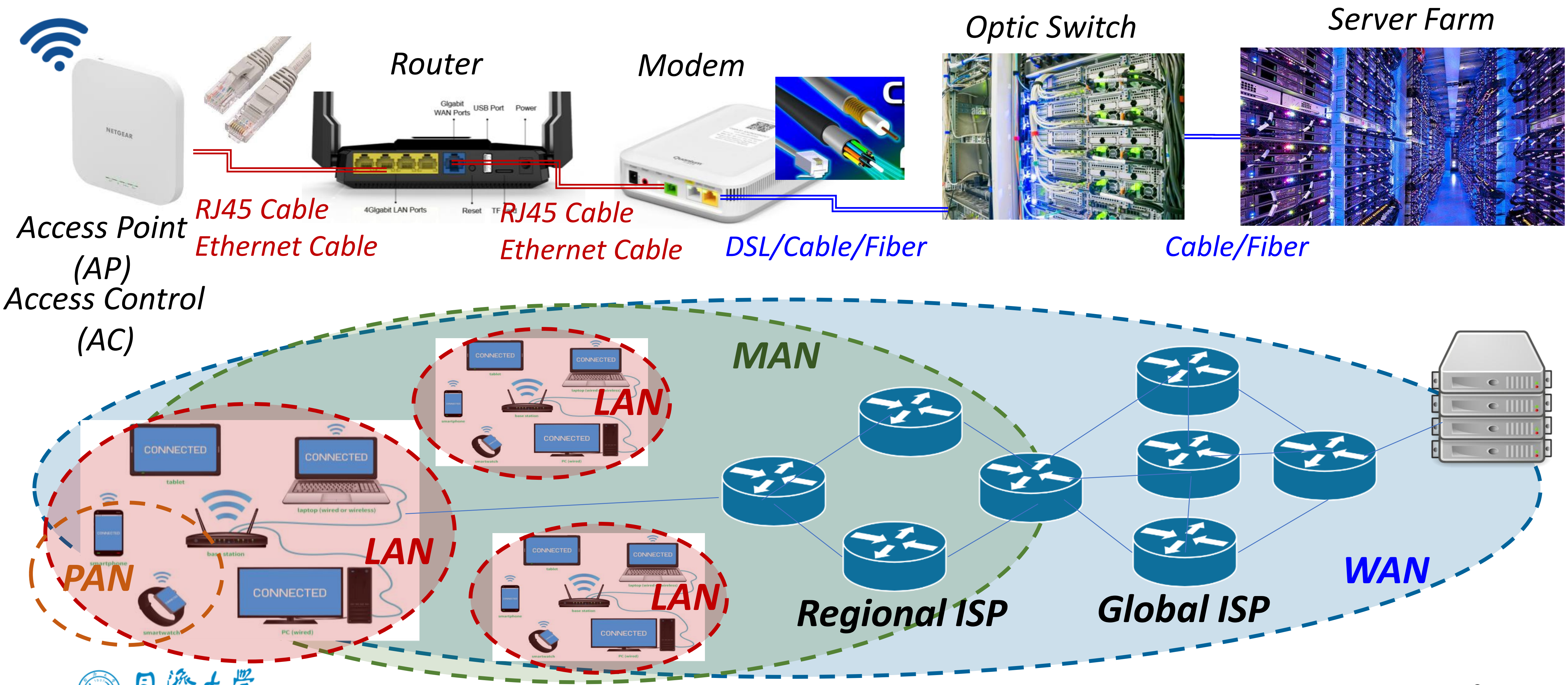
Hello

- ***About me 王洁***
  - ***Homepage:*** *https://faculty.tongji.edu.cn/JieWang/en/index.htm*
  - *Assistant Professor, Tongji University, 2021*
  - *Ph.D. in Computer Engineering from NC State, 2019*
  - *M.S. in Communication Engineering from Tongji, 2013*
  - *B.S. in Communication Engineering from Tongji, 2010*
  - ***Research Interests**: Modeling and Performance Evaluation of Networked Systems (wireless networks, cyber-physical systems, IoT, and social networks), Edge Intelligence…*
- ***About this course 计算机网络***

Connecting to the Internet



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Syllabus

A Brief Introduction (1/2)

- ***A Required Course in Software Engineering, 3 CH, 16 lectures***
- ***Fri.(Every Week) 8:00 – 10:45***
- ***Objectives/ What to learn :***

*A bit of **history**: Why should we learn this course? What is already there?*

*Some **concepts**: What is a computer network? Layered models, protocols, interfaces, services, and applications…*

*The '**beef**': How does every part work? Functions and frequently-used protocols in each layer.*

*A bit of '**engineering**': How to apply the algorithms/mechanisms?*

A Brief Introduction (2/2)

- ***Textbook***
  - *Andrew S. Tanenbaum and David J. Wetherall, Computer Networks (5th Edition), Pearson Education Inc. 2011*
  - *机械工业出版社* **Bottom-up** ‘Protocol Approach’
- ***Other References***
  - *James F. Kurose and Keith W. Ross, Computer Networking, A Top-down Approach (7th Edition), Pearson Education Inc. 2017*
  - *Larry L. Peterson and Bruce S. Davie, Computer Networks, A Systems Approach (6th Edition), Elsevier Inc. 2020*
  - *谢希仁，《计算机网络》，电子工业出版社，2021*

Agenda (Tentative – could change)

Chap.	*Content*	*Lecture*	*Exercise*	*Subtotal*
1	Introduction	5	1	6
2	Physical Layer	5	1	6
3	MAC Sublayer	4	1	5
4	Datalink Sublayer	4	1	5
5	Network Layer	6.5	1.5	8
6	Transport Layer	10	2	11
7	Application Layer	2	1	3
8	Security + Review	3	0	3
Total		***38.5***	***9.5***	***48***

Score and Grades

- **Attendance: 考勤 10%**
- **Homework: 作业 30%**
  - Discussion: 课堂提问、班级群讨论和回答同学问题 6%
  - Individual Homework: 个人作业提交(*3) 24% （**Late submission gets ½ the credit.**）
- **Final Exam: 期末考试 60%**
  - Closed-book, closed-notes 闭卷
- **Academic Integrity: Any breach of academic integrity (including plagiarism, cheating, posting class materials online without permission, and so on) will NOT be tolerated. You will receive ZERO CREDIT for the particular part.** 抄袭等学术不端行为直接记0分
- **Contact：有问题请联系**
  - 王洁 wangjie_tongji@tongji.edu.cn **济事楼** 507
  - Walk-in Office Hour: Tue. 16:00-17:00, or by appointment
  - 通过canvas (推荐使用) 、课程群 (紧急消息) 或邮件 (个人相关) 联系

    邮件标题中注明【CN-26S】并写清学号、姓名等信息

You are expected to … after this course

- *Know why Internet looks like its current status, and how it works*
- *Understand mechanisms in the TCP/IP stack*
- *Write a client-server socket program*

***Or more practically*,**

Pass the final exam! **easy**

Pass an interview when asked about a 3-way handshake, slow-start, throughput, CSMA/CD, CRC, NAT, IP tunneling, etc. **emm…OK**

Configure and troubleshoot your router at home.

Trust me, it is hard!!!

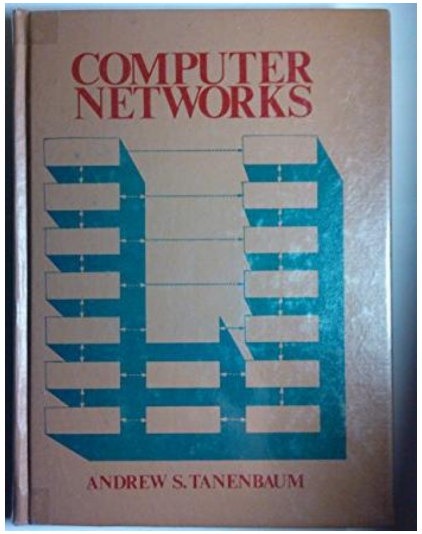
But don't you worry.

- **My English is not good. Will I have difficulty learning this course?**
  - *The textbook is easy to understand. (Sometimes you will even find it verbose)*
  - *High school English is enough to understand the lecture, except…*
  - *Major challenge:* ***new concepts*** *-> figures + explanation*
  - *Q&A in class and on Canvas*

- **Will the exam be hard?**
  - *The exam will be exactly as difficult as the other classes.*
  - *All problems in the exam will have been covered in this class.*

Some Tips on Learning

- ***Complex, but not difficult* -> Build your own *architecture*.**
- ***Key:* *the protocol hierarchy – the software part in networking***



Computer Networks (1st Ed.) by A. Tanenbaum

You will see this figure throughout this semester.

- ***Understand 'WHY', and naturally you will know 'HOW'.***